

Acceptance of Healthcare Robots for the Older Population: Review and Future Directions

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Abstract The rapidly ageing population is placing increasing strain on healthcare services. Robots have been proposed as a way to assist people to stay healthy and safe in their own homes. However, despite the need for such assistive devices and the success of some healthcare robots, other robots have had a poor response. This article reviews the literature about human responses to healthcare robots and summarises the variables that have been found to influence responses. It may be possible to increase acceptance of healthcare robots by properly assessing the needs of the human user and then matching the robot's role, appearance and behaviour to these needs. Because robots have limitations in their abilities, another way to increase acceptance may be to modify the expectations of users to better match robots' abilities. More research needs to investigate potential users' needs and expectations in specific situations and whether interventions to increase the match between robot and human can increase acceptance.

Keywords Older persons · Healthcare · Robots · Acceptance

1 Introduction and Review Strategy

Demographic projections show a rapidly aging population, with the worldwide proportion of people aged over 60 expected to double between 2000 and 2050 [1]. As a result, the number of people living with chronic health conditions is increasing and placing demand on healthcare services. There is also an increasing shortage of healthcare professionals, including caregivers for the elderly [2] and registered nurses [3].

Robots have been proposed as one form of assistive device that can help bridge the widening gap between the need and supply of healthcare services. A robot can be defined as 'a very powerful computer with equally powerful software housed in a mobile body and able to act rationally on its perception of the world around it', [4, p. 9]. Members of the public are most commonly exposed to robots in films such as Star Wars, and in television news about space exploration or other forms of technology. The ideas that people form about robots from these sources may influence how they interact with robots and their acceptance of them in healthcare situations.

A healthcare robot is primarily intended to improve or protect the health and lifestyle of the human user. There are many different types of healthcare robots; from surgical robots [5] to robots that aid with rehabilitation therapy [6], help the disabled and cognitively impaired maintain their independence [7, 8], and motivate people to exercise and lose weight [9], to robots used for telemedicine [10], as well as robots that deliver meals, medication and laundry in hospitals [4].

The aged population presents a group who may particularly benefit from healthcare robots. The term 'aging in place' acknowledges the fact that the majority of older people in western cultures are living independently in their own

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homes, and want to stay there as long as possible [11], thus avoiding the psychologically traumatic and expensive option of moving to a nursing home or hospital away from familiar surroundings, friends and family [12]. Governments and care funders also favour ‘aging in place’ to mitigate the expense of the growing number of aged in care. Researchers are currently trying to develop devices that will enable older people to live comfortably and safely in a home setting. These are generally a less expensive option than home alterations, such as installing a lift. Such devices range from low technology, such as a walking stick, to high technology, such as Japan’s Zojirushi Corp i-pot, which is an electric kettle that will send emails twice a day to the older person’s family letting them know how many times the kettle has been used [13].

While healthcare robots have been criticised as being inferior to human caring and contact [14], often human contact is in short supply, and of poor quality [15]. As much as 45 percent of British caregivers admit some form of abuse of their aged clients [16]. Therefore developing healthcare robots for older people, which aid them to live an inclusive yet independent lifestyle in their own home, may be a useful aid in the care of the elderly.

If healthcare robots are to be successful, they need to be accepted by older people [17]. Acceptance is defined as the healthcare robot being willingly incorporated into the person’s life. For acceptance of robots to occur, there are three basic requirements: a motivation for using the robot, sufficient ease of use, and comfort with the robot physically, cognitively and emotionally. Associated models for acceptance of technology have found that perceived usefulness, ease of use, arousal and pleasure are associated with adoption intentions [18].

This paper reviews the literature on acceptance of healthcare robots by older people. To do this the terms “robotics” and “health” or “aging” or “acceptance” were entered into the databases Psychinfo, IEEEExplore, and Medline.

The review identified a number of healthcare robots that have been developed but not all have been accepted and some have gone out of production due to lack of sales. This review first describes the current status of aged care robots, followed by a summary of the factors found to be associated with acceptance. In some cases the literature search identified technologies that were not purely robotic, or the robots identified had wider applications than just the older population. We included all literature considered relevant to the topic. For clarity, we have summarized the main robot types and their predominant characteristics associated with acceptance in Table 1 at the end of the review.

2 Current Status of Aged Care Robots

A number of research institutes and companies have started to design and test healthcare robots for the elderly. Figure 1 displays some of these robots. Healthcare robots can be broadly categorized into those that provide physical assistance, those that provide companionship, and those that monitor health and safety. Some robotic systems cover more than one of these categories. While some are more directly involved in healthcare provision, others are designed more to provide positive effects through companionship.

An example of a robot more directly involved in healthcare is Pearl, a mobile autonomous robot developed as part of the Nurse-bot project at Carnegie Melon University to aid older people in a rest-home setting [19]. Pearl was designed to guide older people around the rest home and remind them of doctors’ appointments and to take medications. One of the lessons learnt in user-trials was that robots in these settings need to be able to adapt to individual differences, such as hearing ability and walking speeds of different users [20].

The European MOVAID project (1994–1997) investigated a PC and mobile robotic system designed specifically for personal assistance in the home. This robot can be integrated into the older person’s home, helping with basic household tasks and making use of standard appliances easier, for example heating meals in the microwave or removing bed linen [21]. Trials showed that MOVAID can aid independence in the home for disabled users. Care-O-bot is another home care robot that can fetch and carry, give walking assistance, and do some manipulation tasks [22].

AILISA is a French project testing several aspects of healthcare robotics for gerontology, including a smart-shirt for physiological monitoring, a falls monitor and mobility assistance [23]. The research team is developing health smart homes that can incorporate health measures and monitor activities of daily living. Experimental platforms have been installed in geriatric hospital suites in Toulouse and Paris and initial results indicate that the technology has a positive impact but further assessments need to be performed. Ri-man is considered one of the best inventions for medicine in 2006. Ri-man is a robot covered in software tactile sensors that is able to lift a dummy human patient [24]. Ri-man can localize sounds, track human faces, and detect smells.

Social robots such as the Hug [25] have also been designed and tested with the aged, who are often isolated from their families. The Hug has a soft tactile surface that can communicate physical pressure to a remote identical device—so a grandparent can electronically send a hug to a grandchild, and receive one in return.

Some social robots have been shown to benefit the elderly. For example, AIBO the robotic dog, increased communication between dementia patients [26]. AIBO also reduced loneliness in elderly in long-term care facilities [27].



Fig. 1 Examples of healthcare robots. *Top*: Paro (credited to AIST, Japan). *Middle*: MovaID (reprinted from Technology and Disability, 10, Dario P., Guglielmelli E., Laschi C., Teti G. MOVAID: a personal robot in everyday life of disabled and elderly people, 77–93, Copyright (1999), with permission from IOS Press). *Bottom*: Hopis (reproduced with permission of LinuxInsider.com and ECT News Network. Copyright 2003–2009 ECT News Network, Inc. All rights reserved)

Paro is a robotic seal that can respond to visual and aural stimuli and was designed as a pet for those who cannot care for a real animal [28]. The seal is intended to provide similar positive effects to a real pet, including psychological comfort, physiological arousal, and social communication.

It has been tested amongst inpatients and caregivers in several hospital and nursing homes and was launched for sale in the USA in 2008. The i-Cat is another robot that falls under the social robot category and has been tested with older persons [17].

Although many healthcare robots for older people are still in the development and testing phases, some have been commercially produced. Mitsubishi's Wakamaru robot was launched in 2005 as the first commercially available mobile robot designed to help in the home with a communication facility, initially selling 100 for around US\$14500 [29]. Wakamaru can memorise the user's daily activities and send an email to others if the person is unresponsive, has fallen, or if activities are not performed as usual. It can also give reminders for certain activities, such as taking medication. It can recognize ten different people, monitor the house while the person is away, read emails and take messages.

In the context of the older person's home, the division between general assistive robots and health robots can be blurred. Robots that assist with housework, such as the commercially successful Roomba vacuum robot from iRobot (described by Forlizzi, [30]) could be loosely considered healthcare robots if they assist the older person to stay in their own home.

In terms of more healthcare-specific robots, Sanyo launched Hopis the health-monitoring robot in 2004, as the vanguard of the solution to the 'elderly problem' [31]. The talking, toy-like robot is capable of taking blood pressure, body temperature, and blood-glucose levels, diagnosing eye disease, questioning the person about their health, and emailing the details to their doctor. However sales were poor and Sanyo has cancelled production of Hopis [13]. Another commercially produced robot, the Yorisoi Ifbot, designed to be a companion for the elderly, was tested in a rest home but users lost interest after just one month [32]. The lack of uptake of these healthcare robots is likely due to a number of factors.

In order to inform and improve future robot development for the older person, there is a need to study the factors that contribute to the acceptance of robots in healthcare. Failure to conduct and learn from this research will result in the repetition of past mistakes and continued rejection of healthcare robots. A number of studies have investigated how human and robot factors affect responses to healthcare robots. This review attempts to combine these results to form a summary of knowledge and a guide for ongoing research.

3 Individual Factors in the Acceptance of Healthcare Robots

The demographics of the target population should be taken into account when deciding how a robot should be built for

a particular application. In particular the following variables have been found to be important to the acceptance of health-care robots:

3.1 Age

In general, research has found that older age negatively impacts how willing people are to use robots, but this depends on context. When people were asked whether they could imagine living on a daily basis with robots, the most positive responses were from young adults and the least positive from adults over 65 years [33]. However, when asked whether they would accept a robot to help them gain independence when they could no longer handle everyday tasks, acceptance increased particularly in the oldest age range.

Other research has found that when faced with problems, the older old (75+ years) are more likely to give up and accept inconveniences in their home rather than seek assistive solutions, compared with the young-old (65–74 years) [34]. Older people are less comfortable with computers and new technology than younger people [35]. Older people are also more mistrustful and fearful of new technology than the young, and less confident in the abilities of robots [12]. To reduce this anxiety, older people express a preference for robots to be unthreatening, with a female voice, small size, slow-moving and less autonomous, with a serious aspect and single colour. Older people have been found to interact more with a small non-threatening cat robot than young people [36].

3.2 Needs

The needs of the older person strongly affect how they feel towards healthcare robots. Some older people need help with mobility, others with hearing or vision, others need reminders for taking medication and doing other activities, and some need greater social support. Older people have been shown to be prepared to accept assistive devices in healthcare and housing to help maintain their independence [37], especially when there is a perceived need for the device [11]. The failure of some telemedicine projects can be partially attributed to not properly assessing needs [38]. Carefully assessing needs and matching these to the provided technology can result in higher acceptance rates [39]. Expectations of future need can also play a role in acceptance. Higher worry about future loss of cognitive abilities was related to higher evaluations of a robot that could help provide reminders and find things [40].

3.3 Gender

The gender of the user has been found to impact how people react to robots (e.g. [41, 42]). Males asked more questions about the technical aspects of *Paro*, whereas females

focused more on its name [43]. Women have been shown to report higher evaluations of *Paro* in terms of feeling comfortable with it and wanting to interact with it [44]. Siino and Hinds [45] found an interaction between gender and type of job in the way people reacted to the same mobile autonomous hospital robot: engineers and male administrators tended to view the robot as a controllable machine, female administrators and low status staff viewed the robot as a human male, and the nurses saw the robot as un-useful technology. Nomura and colleagues [46] found that males may have stronger psychological connections between anxiety about robot behavior, and negative attitudes about robots' interactions and social influence. Females may have stronger psychological connections between emotional aspects, attitudes about robot interaction, and anxiety toward behavioral aspects.

3.4 Experience with Technology/Robots

Lack of familiarity with technology can be a reason for people feeling uncertain about robots [47]. Direct experience of the usefulness of assistive devices can change older people's attitudes from considering assistive devices unnecessary to considering them very useful [48]. Indeed, the acceptance of devices must be measured over time, as people become familiarized with the robot. Koay et al. [49] found peoples' preferences regarding robots changed over time during a five-week trial as participants habituated to the robot. Over time, they preferred the robot to approach closer to them, and differences in responses to robot appearances reduced. Over a two-month study, residents in a care facility developed personal relationships with *Paro*, which then resulted in stronger social ties amongst the residents [50].

3.5 Cognitive Ability and Education

There is only a small amount of research on the effects of level of education on attitudes towards robots. An Italian study of 123 older people, found that higher education was associated with greater acceptance of technological solutions for everyday problems [34]. Libin and Libin [36] found that older dementia patients' level of cognitive ability predicted the degree of use of a robotic cat. While the higher functioning patients played with the robot; the more cognitively impaired patients played equally as much with a non-robotic plush toy cat. There was no correlation between level of satisfaction and education in patients who had undergone robotically-assisted coronary artery bypass grafting, although the sample size of 20 was small [51].

3.6 Culture

Cultural differences have also been found in attitudes towards robots. Americans were more positive than other cul-

tures in their attitudes towards robots [52]. American participants also liked touching a therapeutic robotic cat NeCoRo and liked the cat looking at them more than Japanese participants did [36]. Differences also exist between European groups, for example French-speaking people were found to be more accepting of a human-looking robot than Germans [33].

A cross-cultural study of people's assumptions about humanoid and animal type robots found several differences between Japan, Korea and the USA [53]. Notably, Japanese respondents thought that humanoid robots were more capable of emotions, could be considered more like humans than tools, and believed more strongly that robots could fulfill a communication role in the home, compared to the other ethnic groups. Japanese also thought robots were more likely to perform nursing, education and social roles whereas Koreans thought robots were more likely to play a role in emergency hospital situations and Americans thought robots were more likely to be used in roles that were difficult for humans to fulfill. US participants had fewer reservations about ethical issues. The reasons behind these cultural differences may reflect differences in belief systems and motivations, but may also reflect experience and exposure to robots in films and in daily life.

A cross-cultural study of reactions to Paro showed that participants from all the countries in the study (UK, Sweden, Italy, Japan, Korea, Brunei) evaluated Paro highly. Desire to talk to Paro was lower in the UK, Sweden and Italy, and this was thought to be because the robot did not understand those languages [44].

Cultural differences in attitudes towards aging and independent living, as well as differences in how societies traditionally care for older people will likely result in cultural differences in the acceptance of robots, and this is an area for future research.

3.7 Role

Peoples' roles affect how they respond to robots. Preliminary research has shown that reactions of staff to healthcare robots can be mixed. Wasen showed that assistant surgeons sometimes feel robotic surgery isolates them from the head robotic surgeon, and are annoyed that the bulky surgical robots can be difficult to move around during surgery [54]. A study of nurses' reactions to an automated medication dispensing system showed that many nurses were distrustful of the technology, they had unrealistic expectations, and felt that their job security was threatened [55]. Nurses in rest homes can feel stressed and some researchers are aware of assessing the effects of introducing a robot on these stress levels [56].

3.8 Anxiety and Attitudes Towards Robots

There has not been much research into the influence of anxiety and attitudes towards robots on acceptance, particularly with regards to healthcare robots and older people. Heerink et al. [57] found older people accepted robots more readily if they perceived that they would enjoy the interaction more. Nomura et al. [46] found that negative attitudes and anxiety about robots were related to delays in interaction behaviour with a human-like robot who asked, "Tell me one thing that recently happened to you". The results suggested that people with high negative attitudes and anxiety about robot interactions tended to talk more about emotional subjects with the robot. An investigation into attitudes towards personal service robots found that people do hold opinions about service robots even though such robots are not yet fully realized; most would be willing to accept one into their own home or can see their uses for the elderly or disabled [58].

This work suggests that emotions and attitudes towards healthcare robots will influence reactions, but further work is needed in this area.

4 Robot Factors in the Acceptance of Healthcare Robots

When building a robot, there are many different variables to consider. Several studies have investigated how different robot factors are involved in robot acceptance, and these are described below.

4.1 Appearance

A large number and variety of robot factors can influence responses to robots. Most previous research has focused on the robot's appearance, rather than its behaviour, which is a topic for future study. The appearance of products is very important as it contributes to the elder's sense of identity [30]. Careful consideration of the health robot design is needed to minimise the stigma of disability. Older people value not only independence, but also the appearance of independence, and they may not use assistive robotic devices that they feel portray them as disabled, dependent, weak or feeble [59].

The robot's appearance influences how people appraise the abilities of the robot [60]. If the robot looks 'high-tech' people expect it to perform at a high level. Keeping people's expectations low was one of the main reasons Breazeal's [61] social robot 'Kismet' was given the appearance and behaviour of a young creature—on the premise that if the robot has an adult humanoid appearance people will expect it to be able to converse naturally; which requires very complex software and programming. However the look of a 'young

creature' is probably unlikely to inspire confidence for the role of a healthcare robot. This may have been one of the reasons for the lack of interest in the health robot Hopis, which has a toy-like appearance.

The different aspects of a robot's appearance affect the human user in a variety of ways, as is further described below.

4.2 Humanness

There is debate as to how human a robot should look, with many preferring a less humanlike appearance. The 'uncanny valley' refers to peoples' unease when a robot looks almost—but not quite—human [62]. Of 2000 people surveyed at a Swiss Expo, only 19% said they would prefer a human looking robot, compared with 47% who said they would not [33].

Oestreicher surveyed a small sample of social science students to discover important aspects of appearance and behaviour for robots to be socially acceptable [63]. The results were not analysed with appropriate statistics, but basic visual inspection of scores suggested that robots should be more machine-like than human-like, and they should be easy to use, safe, and reliable. Results also suggested that it is not so important for robots to give facial expressions, nor to have a personality.

Different applications may require different degrees of human-likeness. Panels of teachers, therapists and family members recommended that a robotic toy for autistic children should not be too human-like, but more mechanical in appearance, with a degree of familiarity, and the child should be able to exercise a degree of choice and control over the robot via a set of buttons [64]. Initial work with autistic children indicates that they interacted more with a featureless robot toy than with a prettier, more human version [65].

The ROBOCARE Domestic Environment aims to provide cognitive support to a person [40]. Movies of the robot interacting in several different scenarios were shown to people aged 56–84, with the participants randomly assigned to see either a robot with a face or without. Results showed that people preferred the robot without a face.

A health robot need not look human at all, but non-humanoid robots raise different issues. People who are less fond of animals have a mixed response towards a robot that has an animal appearance [28, 36]. People with a general dislike of pets or animals tended not to engage with the robotic animal. However people who were fearful of risky aspects of real animals, such as being bitten, or being unhygienic, did tend to engage with the robots as they felt the robotic animal was safe.

Research has found that a human-like appearance can encourage humans to share responsibility with a robot [66], so this may be appropriate in job-sharing roles.

4.3 Facial Dimensions and Expressions

Just as people react to facial expressions on other people, they make sense of and react to expressions on a robot's face [67]. However, the facial expressions people expect on a robot in a typical interaction may not correspond with Ekman's universal expressions [68], with 'smile' and 'calm' replacing 'disgust' and 'surprise' [69].

The dimensions of a robot's head can affect how people perceive the robot, with a shorter chin associated with perceptions of higher sociability and higher intentions to follow the robot's medical advice [60]. Large smiles that have slow transitions are seen as more appealing than small smiles or those with quick transitions or that are static [70].

4.4 Size

The size of the robot is often important for practical and role reasons. Breazeal made the social robot Kismet small to encourage a caring, protective response from humans [71]. The surgical robots Zeus and da Vinci are large, but can sometimes get in the way of the assistant surgeons [54]. Most older people have expressed a preference for a small robot in a home setting [34].

4.5 Gender

The apparent gender of the robot can affect how people react to it. For example, people explained dating norms more fully to a male robot than a female robot [71]. It is thought that the perceived ability and knowledge of the robot will be inferred by traditional role stereotypes and thereby influence how much people tell the robot about a related subject.

4.6 Personality

Just like a human's personality, a robot's personality will affect how people react. People preferred a robot whose personality was matched to their own in terms of extraversion [72]. Making home robots more socially intelligent can contribute to acceptance [73]. A caring and empathic robot personality has been shown to encourage interaction [74]. Other research shows that people are more comfortable interacting with a more socially communicative robot [17]. The voice and appearance of the robot can affect how people rate the robot's knowledge and sociability, which in turn can affect intention to follow its advice [60].

Some evidence suggests that the personality of a robot needs to be matched to its role. A robot designed to encourage people to exercise was programmed with two different personalities [75]. While the participants reported liking the fun robot more, they exercised more when the robot displayed the serious personality.

4.7 Adaptability

Research suggests that a robot's ability to adapt its behaviour to the user's preferences and personality can improve acceptability. A robot designed to assist with post-stroke therapy could successfully adapt its speed and interaction style on-line to match the preferences of the individual user [76]. Trials of Pearl with older people discovered that the robot needed to be able to adapt to the individual [20].

Robot adaptability is likely to be very important for older people, who will have a range of individual problems, such as poor eyesight, limited movement, and impaired hearing. Individuals' abilities are likely to further decline over time, and the robot will need to adapt to these changes. A system has been proposed for an adaptive pet-type robot for healthcare tasks for chronically ill, disabled and older users [77].

5 Discussion

Based on the results of the review, several issues are discussed and recommendations are made for robotic design and future research.

5.1 Characteristics of Healthcare Robots for Older People

Table 1 lists the robots included in the review that were either designed for the care of older people or investigated for their benefits with older people. The robots are categorized into physical assistance, safety/medical monitoring, and social companion types. Identifiable characteristics important to acceptance have been summarized for each robot, as well as the country the robot is from.

The global nature of the developments is clear, spanning Asia, Europe and America. There is a mix of machine-like, human-like, and animal-like robots. Many of the human-like or animal-like robots have become commercial products in Japan, especially animal-like robots built for social companionship. Some of these, for example Paro and Aibo, have been used successfully with older people both within and outside Japan, demonstrating their cross-cultural application. Animal-like fluffy robots designed more for healthcare tasks have been less popular, for example Hopis. More human-like robots designed for companionship have also been unpopular, for example Ifbot. This suggests a good match between animal robots and companionship. More technically difficult machine-like and human-like robots designed for physical assistance are still in the research and development phases. Most robots have little in their appearance to specify their gender, although some are specified as male or female by the makers, while voice characteristics can sometimes be chosen by the user.

5.2 Robot Design

Research into the factors that affect responses to healthcare robots for the elderly is still in its infancy. Many studies have been artificial and observational rather than experimental and in real settings. However, early findings suggest that the robot must meet the person's needs, be slow, safe and reliable, small, easy to use and have an appearance that is serious, not too human-like, not patronizing or stigmatizing, and have a serious personality. There is not enough research to advise robot gender. Further studies are required to test appearance and behaviour requirements for specific applications. Pre-consultation with potential users of a healthcare robot about needs and preferences is important, but expressed preferences are likely to change with changing needs, training and experience with a robot.

5.3 Measures

There are well-validated measures to assess elderly people's needs (see [78]), but there are comparatively few validated measures to assess needs and preferences for healthcare robots. There are also few validated measures of responses and acceptance of robots. Many different outcomes of robot deployment can be assessed, including safety, impact on care, quality of life, and user satisfaction [79]. Measures of response may include frequency of use, user-ratings and behavioural responses.

Responses to hospital robots are starting to be assessed. For example a mobile robotic telepresence was shown to reduce patients' post-operative hospital bed-stays (resulting in substantial savings per patient) however there was no analysis of how this occurred [80]. A study assessing satisfaction with a robot 'telerounding' post-surgery focused on satisfaction with hospital care in general rather than with the robot itself [81]. Although one study has tried to assess patients' perceptions of surgical robots, these were only superficially assessed [51]. Ad hoc or adapted measures have often been used to assess responses to robots, but there is usually little or no information offered on how they were constructed or on their validity, and these have often been used in small samples. The Person-Robot Complex Interactive Scale (PRCIS) was developed for use with NeCoRo, a robotic cat, although no measures of validity or reliability were presented [82]. A promising new questionnaire has recently been published assessing five concepts around users' perceptions of robots [83].

Some studies have borrowed measures from social psychology to assess response to robots, such as the Working Alliance Inventory [84] or the Dyadic Interaction System [85, 86]. Further publication and use of psychometrically validated scales for responses to healthcare robots will contribute to the field.

Table 1 Robots designed for or tested with older people categorized by robot type

Robot name [reference]	Short description (humanness, facial expressions, size, gender, role)	Country	Commercial product or research development
<i>Providing physical assistance</i>			
MOVAID [21]	Machine-like, no head/face, approx. 2 metres, no gender, heating/delivering food, changing bed linen, kitchen bench cleaning	Italy	Research development
Care-O-bot [22]	Human-like, head with static face, no gender (female voice), approx. 1.5 metres, walking aid and fetch and carry task	Germany	Research development
AILISA [23] (robot MONIMAD)	Machine-like, no head/face, approx. 1 metre, no gender, ambulation and transfer assistance	France	Research development
Ri-man [24]	Human-like, head with static face, approx. 1.5 metres, no gender (male voice), carrying people	Japan	Research development
<i>Safety/monitoring</i>			
Pearl [19]	Human-like, head capable of facial expressions, approx. 1 metre, female, reminders for daily activities, guiding around rest home	USA	Research development
Wakamaru [29]	Human-like, head capable of facial expressions, approx. 1 metre, male (voice sometimes female), home security, managing schedules, information service, reporting unusual situations	Japan	Commercial product
Hopis [31]	Dog-like, fluffy, head has static face, approx. 0.3 metres, no gender, measuring blood pressure, blood sugar and temperature	Japan	Commercial product
<i>Social companion/communication robots</i>			
Hug [25]	Cushion-like, soft, no head/face, approx. 0.5 metres, no gender, social communication	USA	Research development
Ifbot [32]	Human-like, head has static face, approx. 0.45 metres, no gender, social communication, mental stimulation, entertainment	Japan	Commercial product
NeCoRo [36]	Cat-like, fluffy, head capable of facial expressions, approx. 0.25 metres, no gender, social companionship	Japan	Commercial product
Paro [28]	Seal-like, fluffy, capable of facial expressions, approx. 0.25 metres, no gender, social companionship	Japan	Commercial product
iCat [17]	Cat-like, hard surface, capable of facial expressions, 0.38 metres, no gender, social companionship	The Netherlands	Research development

Where multiple measures have been used to assess responses e.g. observation, self-report and EEG readings [28] a discrepancy may be observed between what people say and what they do. For example, one man spoke negatively about Paro, which was in marked contrast to his happy, interactive behaviour with the seal robot, and his improved corti-

cal neuron activity. The authors proposed the older man's expressed dislike for the seal robot was caused by embarrassment. Conflicting responses to health robots can make it difficult to accurately assess responses, particularly if only self-reports are used. Measurement of more than one type of response may improve assessment.

5.4 User Control

Considering the number and variety of variables that may affect responses, allowing the user to personalise the robot to their own needs may be one way of accommodating individual differences. An example is the pediatric story telling robot, PETS, which has been beneficial for both physical and emotional therapy [87]. Children can select the robot's appearance by choosing from a variety of animal robot parts and snap them on to the body of the robot.

It may be useful to allow the user to choose the gender of the robot, or its personality. As well as achieving congruency with the users' gender-role stereotypes and preferences, this could also help give the older person a sense of personal autonomy and control over the robot and their own health. This sense of independence is often compromised, and is very important for many older people [11].

5.5 Modifying User Expectations

There are limitations on the extent to which a robot's appearance and especially its behaviour can be modified to meet human expectations and desires. Limitations to current technology prevent some modifications, for example, only very limited intelligence and common sense reasoning can be provided in practical robots at present. Limitations also apply to the robot's sensing and motor abilities. A promising and achievable strategy may be to carefully understand and modify human expectations to lie within a robot's possible range of appearance and behaviour. However, this requires a high degree of understanding about how robot appearance and behaviour interacts with human expectations and attitudes.

5.6 Conclusion

Robots have great potential to aid independence and improve health outcomes in the aged, and relieve the burden for caregivers. A number of healthcare robots have been developed, however, not all have been successful. This review has identified the importance of considering the perspectives from a range of stakeholders, such as the older person, their family and medical staff, and carefully assessing their expectations and needs. A number of individual variables influence responses to health robots, and these include age, needs, gender, cognitive ability, education level, past experience, and culture. Robot variables too, are central, and include appearance (humanness, and facial characteristics), size, gender, ergonomics, role and 'personality'. These factors align with the findings of a recent systematic review of factors in the acceptance of user technology [88]. Ten factors were found that were categorised under three headings: organizational factors (voluntariness, task profession), technological factors (individual/group, complexity, purpose) and

individual factors (age, gender, experience, cultural background, intellectual capability). They also fit with a recent paper applying social psychology to the acceptance of domestic robots [89].

A single perfect design of a healthcare robot is unlikely, and carefully assessing individual needs and preferences and matching these to the robot may enable greater acceptance. This may also allow the user to feel that their robot is less stigmatizing, and allow the robot's functionality to change over time. After experiencing the robot, the user may wish to add functionality or change their preferences, gradually increasing acceptance and use. This 'user matching' strategy aims to alter the robot to match the human's expectations. While matching will solve some of the problems, more research is needed to better understand the variables that influence acceptance of specific robot types. This will aid the design of more generic models for multiple users in group situations, such as rest-homes and hospitals. A series of small scale trials will need to be performed to define optimal robot parameters in specific settings. These results can then be built into robot designs and larger trials performed.

An alternative approach is to alter users' expectations to match the capabilities of the robot. This approach offers more flexibility and potentially more success as it decreases the effort required to cater for each individual through robot modifications and helps control for current engineering limitations. Few studies in robotics have focused on the role of existing expectations, and this area needs further investigation. If research can identify the expectations that influence the quality of interactions, then these expectations can provide a framework for interventions to target and alter. In general, people are not familiar with robotic healthcare, so setting up appropriate expectations may be both feasible and effective. Once the role of expectations has been established, intervention studies can be tested. Further development and validation of psychometric measures of responses to robots will contribute to the rigour by which these interventions can be assessed.

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